



Xi'an Jiaotong-Liverpool University

西交利物浦大學

RBE211TC DYNAMIC SYSTEMS

VIBRATION OF MECHANICAL SYSTEMS – SDOF SYSTEMS (PART 2)

Dr Sze-Hong Teh

E-mail: Szehong.Teh@xjtlu.edu.cn



Lecture Outline

1. Physical meaning of resonance
2. Effect of damping on vibration response
3. Behaviour below, near, and above resonance
4. Why operating region matters
5. Basic idea of vibration isolation
6. Engineering interpretation and applications



Recap: Forced Vibration of SDOF Systems

Previously, we introduced:

- harmonic excitation
- transient and steady-state response
- frequency ratio $r = \omega/\omega_n$
- amplitude ratio
- resonance
- effect of damping on amplitude and phase



From Calculation to Interpretation

In vibration analysis, it is not enough to calculate:

- natural frequency
- damping ratio
- amplitude ratio

We must also interpret:

- whether the system is operating safely
- whether resonance is likely
- whether damping is sufficient
- whether vibration should be reduced or exploited



Physical Meaning of Resonance

Resonance occurs when:

$$\omega \approx \omega_n$$

Physical interpretation:

- the excitation matches the natural tendency of the system to vibrate
- energy is transferred very efficiently into the motion
- the response amplitude becomes large



Energy Viewpoint of Vibration

In a vibrating mechanical system:

- inertia and stiffness exchange energy
- damping dissipates energy
- external forcing supplies energy

Near resonance:

- supplied energy accumulates efficiently in the motion
- response becomes more severe unless damping limits it



Why Is Resonance Often Undesirable?

Severe resonance may lead to:

- large displacement
- large stress and fatigue
- poor motion accuracy
- excessive noise
- shortened component life
- structural or mechanical failure



Failed roller bearing due to fatigue.



Resonant oscillations of the Tacoma Narrows bridge.



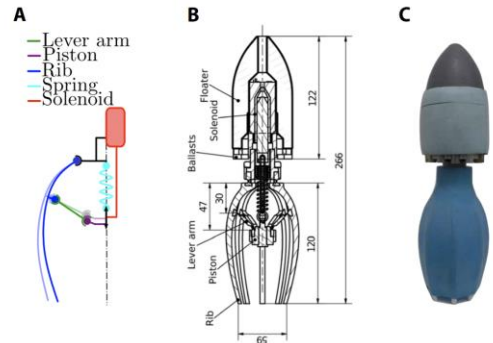
Can Resonance Ever Be Useful?

In some applications, resonance can be exploited to:

- amplify motion
- reduce required actuation effort
- enhance sensitivity
- improve energy transfer

Examples:

- resonant sensors
- energy harvesters
- certain robotic actuating mechanisms



Flexible Bioinspired Resonant Robot
(Bujard et al.)



Examples of Resonance in Robotics and Engineering

Resonance-related behaviour may appear in:

- robot links and flexible arms
- mobile robots on uneven terrain
- rotating machinery
- suspension systems
- lightweight structures and frames

Key point:

Understanding resonance helps engineers predict and manage system behaviour.



Recap: Why Damping Matters

Damping:

- reduces response amplitude
- limits the resonance peak
- changes the phase relationship
- helps control vibration severity

Key idea:

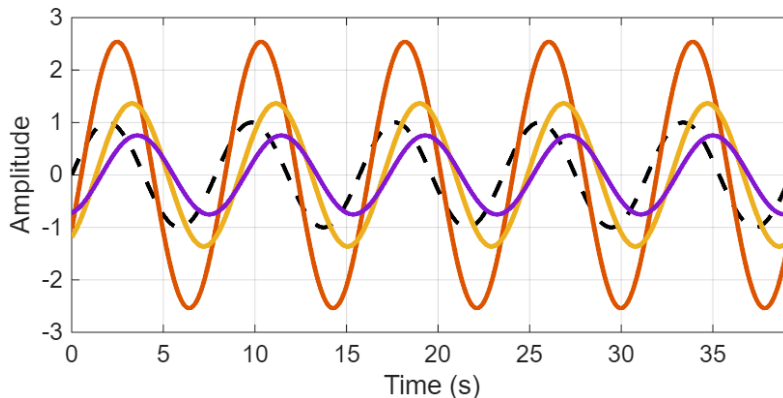
Damping does not remove vibration completely, but it helps make the response more manageable.



How Does Damping Change the Response?

Compared with a lightly damped system, a more heavily damped system generally has:

- lower vibration amplitude near resonance
- less severe resonance peak
- more controlled response
- different phase relationship with the excitation



Input force and harmonic responses when $r = 0.8$ for different ζ .
(Dashed: Input, Red: $\zeta = 0.1$, Orange: $\zeta = 0.4$, Purple: $\zeta = 0.8$)



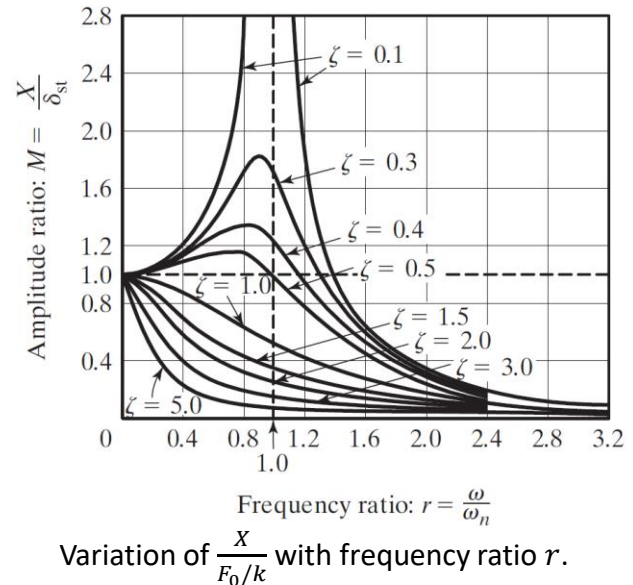
Effect of Damping on Response Amplitude

For forced vibration:

- low damping allows strong amplification near resonance
- higher damping reduces the maximum response
- damping helps prevent excessively large vibration

Key point:

Damping limits the severity of resonance.



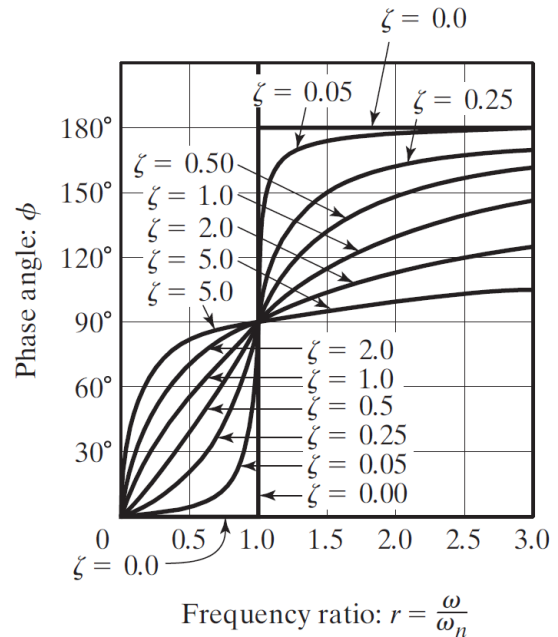
Effect of Damping on Phase Relationship

Damping also affects the phase lag between input and response:

- at low frequency, response is nearly in phase with excitation
- near resonance, phase lag increases significantly
- at high frequency, response lags further behind

Key point:

Damping affects both amplitude and timing of motion.



Variation of ϕ with frequency ratio r .



Does Damping Remove Resonance?

Even with damping:

- resonance effects may still occur
- response amplitude may still become large near $\omega \approx \omega_n$
- the resonance peak is reduced, not removed completely

Key point:

Damping reduces resonance severity, but does not make resonance irrelevant.



Damping as an Engineering Trade-Off

Increasing damping may:

- reduce vibration amplitude
- improve robustness near resonance
- reduce oscillation severity

But it may also:

- increase energy dissipation
- reduce responsiveness in some systems
- affect efficiency or performance



Behaviour Below Resonance

When

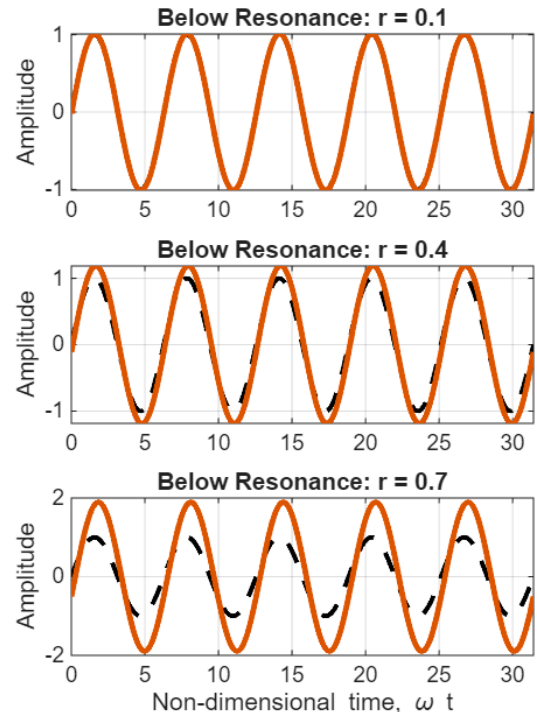
$$\omega < \omega_n$$

the system typically shows:

- moderate response amplitude
- motion that follows the excitation more closely
- lower risk than near resonance

Physical interpretation:

the system can respond relatively easily to the forcing.



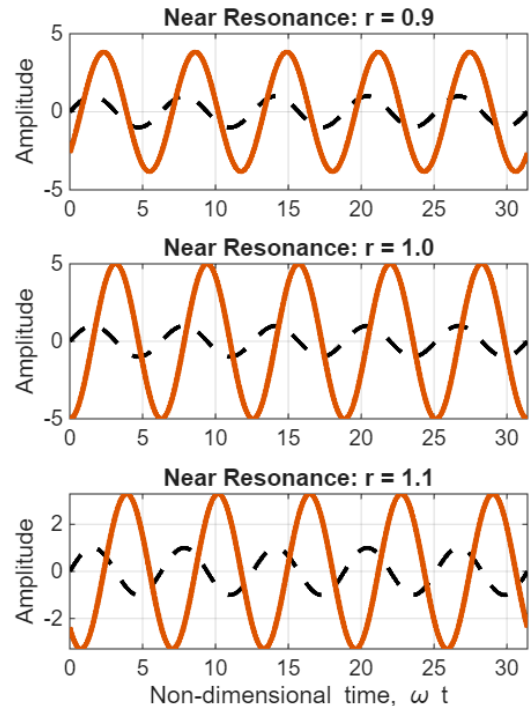
Behaviour Near Resonance

When

$$\omega \approx \omega_n$$

the system typically shows:

- large vibration amplitude
- efficient energy transfer into vibration
- greatest sensitivity to damping
- highest risk of severe vibration



Behaviour Above Resonance

When

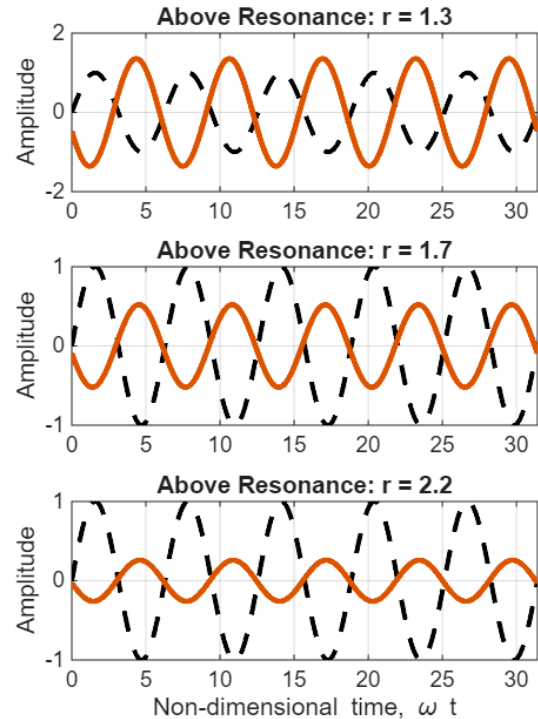
$$\omega > \omega_n$$

the system typically shows:

- reduced response amplitude compared with resonance
- motion is less able to follow the excitation
- increasing phase lag

Physical interpretation:

The system cannot respond effectively to very rapid forcing.

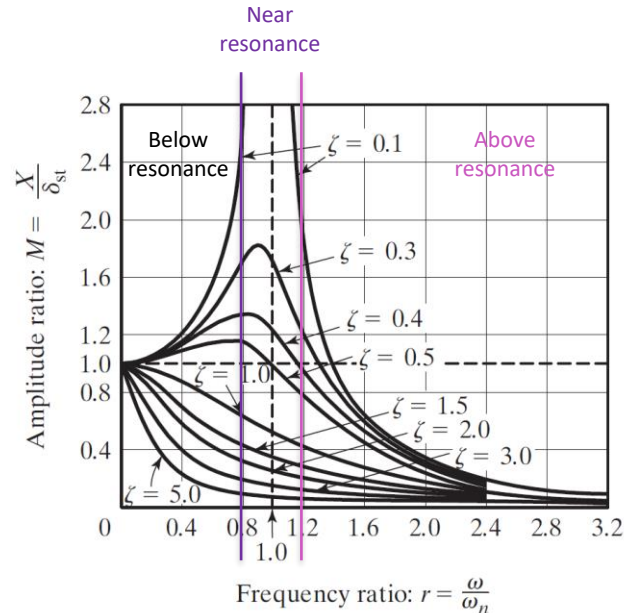


Why Does the Operating Region Matter?

The same system may behave very differently depending on excitation frequency:

- below resonance → usually moderate response
- near resonance → severe response possible
- above resonance → response often reduced

Engineering implication:
Safe and effective operation
requires awareness of frequency
region.



Variation of $\frac{X}{F_0/k}$ with frequency ratio r .



Selecting Damping to Limit Resonance Amplitude

Example 1

A damped SDOF system must operate near resonance. To ensure acceptable vibration behaviour, the maximum magnification factor must satisfy

$$M_{\max} \leq 3$$

Estimate the minimum damping ratio required. Use

$$M_{\max} = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$



Selecting Damping to Limit Resonance Amplitude

Example 1



Selecting Damping to Limit Resonance Amplitude

Example 1



Selecting Damping to Limit Resonance Amplitude

Example 1



Why Is Vibration Isolation Important?

In many practical systems, we want to:

- reduce vibration transmitted from a machine to its support
- reduce vibration transmitted from a vibrating base to a system
- protect precision equipment
- improve comfort, safety, and durability

Typical applications:

- machine mounts
- vehicle suspensions
- instrument supports
- robotic systems on moving platforms



Vibration isolators used in industry.

Two Common Vibration Isolation Situations

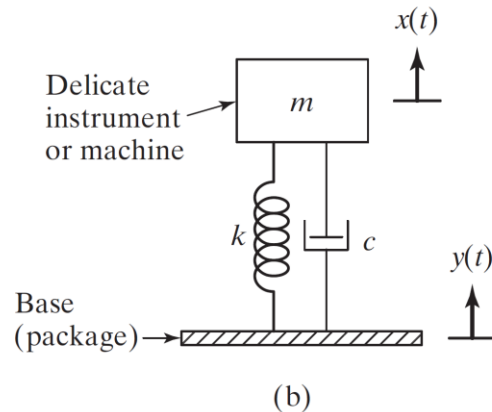
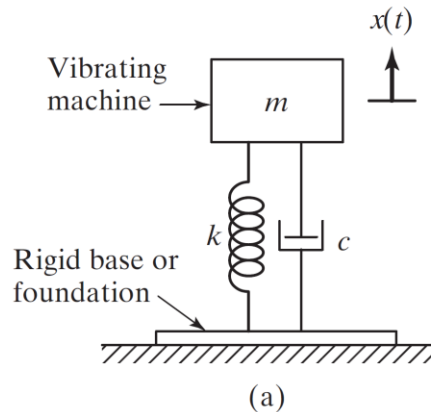
Vibration isolation is commonly considered in two ways:

(a) Isolation of the foundation from the machine

- reduce force transmitted to the support

(b) Isolation of the system from base motion

- reduce motion transmitted from the support to the system



Vibration isolation.



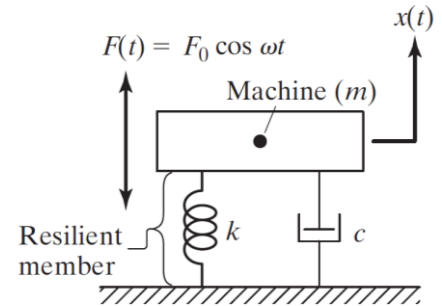
Isolation Model for a Vibrating Machine

A simple isolation model consists of:

- machine mass m
- spring stiffness k
- damping coefficient c
- harmonic exciting force $F(t)$

Goal:

- Reduce the force transmitted to the foundation



Foundation or base
Machine and resilient member on rigid foundation.



Force Transmissibility

Define the force transmissibility as

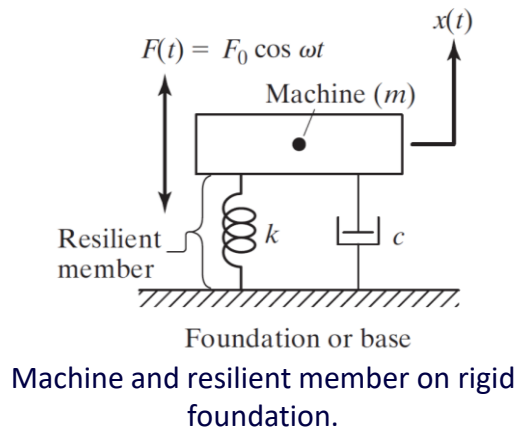
$$T_f = \frac{F_T}{F_0}$$

where:

- F_T : amplitude of transmitted force
- F_0 : amplitude of exciting force

Interpretation:

- $T_f < 1$: isolation achieved
- $T_f > 1$: transmitted force amplified



Force Transmissibility

For a damped system, the force transmissibility is

$$T_f = \sqrt{\frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}}$$

where:

- $r = \frac{\omega}{\omega_n}$
- ζ : damping ratio



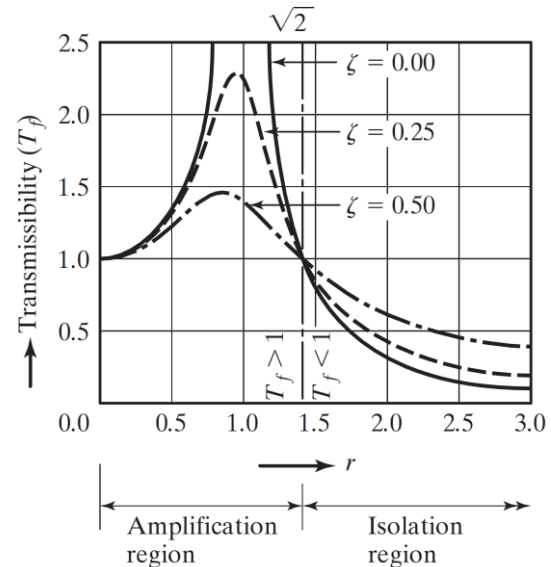
When Is Isolation Achieved?

From the transmissibility curve:

- for low frequency ratios, transmitted force may be large
- near resonance, transmissibility may be high
- isolation is achieved when transmissibility becomes less than 1

Key point:

Effective isolation generally requires operation sufficiently above resonance.



Variation of transmission ratio T_f with r .



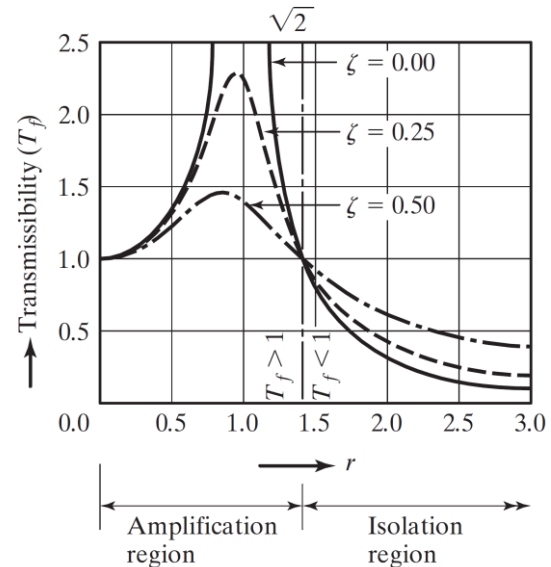
How Does Damping Affect Isolation?

Damping has two important effects:

- near resonance, it reduces the transmissibility peak
- in the high-frequency isolation region, it may reduce isolation performance

Key point:

Damping is beneficial near resonance, but may not always improve isolation at high frequency.



Variation of transmission ratio T_f with r .

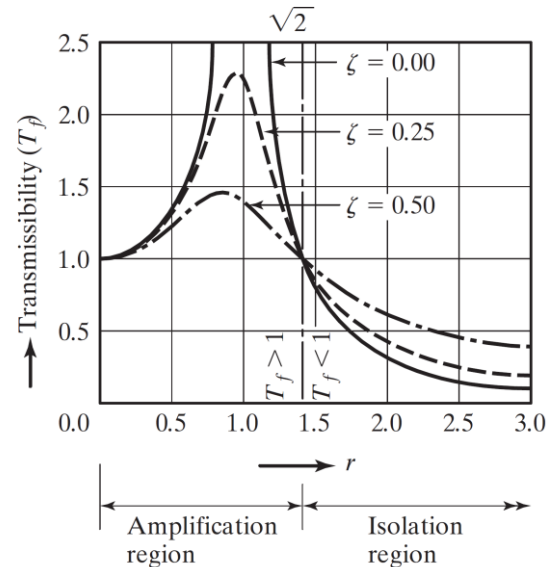


Physical Interpretation of Isolation Behaviour

- **Below resonance:** system tends to follow the excitation strongly
- **Near resonance:** vibration and transmitted force can become severe
- **Above resonance:** transmitted effect reduces and isolation becomes possible

Engineering lesson:

An isolator must be designed together with the expected operating frequency range.



Variation of transmission ratio T_f with r .



Checking Whether Isolation is Achieved

Example 2

A machine is mounted on an isolator with:

- damping ratio $\zeta = 0.1$
- frequency ratio $r = 2.0$

Determine the force transmissibility using

$$T_f = \sqrt{\frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}}$$

Comment on whether vibration isolation is achieved.



Checking Whether Isolation is Achieved

Example 2



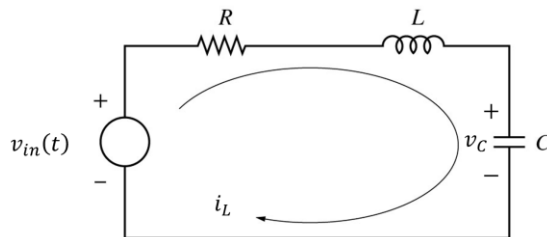
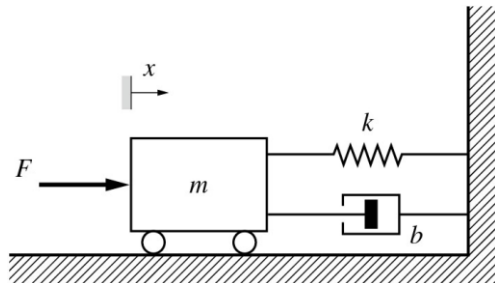
Link to Electrical Second-Order Systems

Mechanical vibration behaviour has strong analogies with electrical second-order systems:

- resonance behaviour
- damping effect
- frequency-dependent response
- response amplification near characteristic frequency

Connection to Lab 4:

- RLC systems also show second-order dynamic behaviour



Review

1. Physical meaning of resonance
2. Effect of damping on vibration response
3. Behaviour below, near, and above resonance
4. Damping as a design trade-off
5. Vibration isolation and transmissibility
6. Engineering interpretation of vibration phenomena

